Care Transition Efficiency & Placement Outcome Analytics

Abstract

This project analyzes operational efficiency and placement outcomes within the Unaccompanied Alien Children (UAC) Program. The study evaluates how effectively children transition from CBP custody to HHS care and eventually to sponsor reunification. Key metrics such as Transfer Efficiency Ratio, Discharge Effectiveness, Pipeline Throughput Rate, and Backlog Indicators were analyzed using time-series analytics. The findings provide insights into operational bottlenecks, reunification delays, and care pipeline performance.

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1. Introduction

The UAC Program functions as a multi-stage humanitarian care pipeline involving apprehension, CBP custody, HHS sheltering, medical support, case management, and sponsor reunification. Beyond system capacity, the efficiency and reliability of transitions between stages are critical to child welfare outcomes.

This project focuses on evaluating care transition efficiency and identifying process bottlenecks using operational analytics.

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2. Objectives

Primary Objectives

- Measure CBP to HHS transfer efficiency
- Evaluate discharge and sponsor placement outcomes
- Identify process delays and bottlenecks

Secondary Objectives

- Support faster reunification
- Improve case-management workflows
- Enable policy-level operational improvements

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3. Dataset Description

The dataset contains daily operational records from 2023–2025.

Column	Description
Date	Reporting date
Children apprehended and placed in CBP custody	Daily intake volume
Children in CBP custody	Active CBP care load
Children transferred out of CBP custody	Transfers into HHS
Children in HHS Care	Active HHS care load
Children discharged from HHS Care	Sponsor placements

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4. Methodology

The dataset was processed using Python and Pandas. The Date column was converted into datetime format and arranged chronologically.

Several process metrics were calculated:

- Transfer Efficiency Ratio = Transfers ÷ CBP Custody
- Discharge Effectiveness = Discharges ÷ HHS Care
- Pipeline Throughput Rate = Total Exits ÷ Total Entries
- Backlog Indicator = Transfers − Discharges

Trend analysis, weekday comparisons, and backlog detection methods were applied to evaluate operational efficiency.

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5. Results and Findings

The analysis identified fluctuations in transfer efficiency and discharge effectiveness across the reporting period.

Periods of low discharge effectiveness contributed to backlog accumulation and increased shelter burden. Sustained positive backlog indicators suggested unresolved case growth and operational stress.

Weekday analysis showed improved discharge performance during standard operational weekdays compared to weekends.

Pipeline throughput decreased during high-intake periods, indicating reduced system efficiency under elevated demand conditions.

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6. Discussion

The findings demonstrate that care pipeline performance depends heavily on the speed and consistency of transitions between operational stages.

Reduced transfer efficiency may indicate shelter constraints or administrative delays, while lower discharge effectiveness can increase unresolved case accumulation and prolong shelter stays.

The backlog indicator proved useful for identifying periods where inflows consistently exceeded successful exits, highlighting operational strain and reduced reunification efficiency.

Overall, the study emphasizes the importance of real-time operational analytics in improving humanitarian care delivery and process transparency.

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7. Recommendations

- Develop real-time operational dashboards
- Implement early-warning bottleneck detection systems
- Improve sponsor placement workflows
- Increase staffing during high-backlog periods
- Strengthen weekend operational capacity

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8. Conclusion

This project reframed the UAC dataset from a simple capacity monitoring perspective to a process efficiency and placement outcome evaluation framework. By analyzing transfer efficiency, discharge effectiveness, throughput performance, and backlog accumulation, the study provided actionable insights for improving reunification timelines, reducing delays, and enhancing humanitarian response operations.

References

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